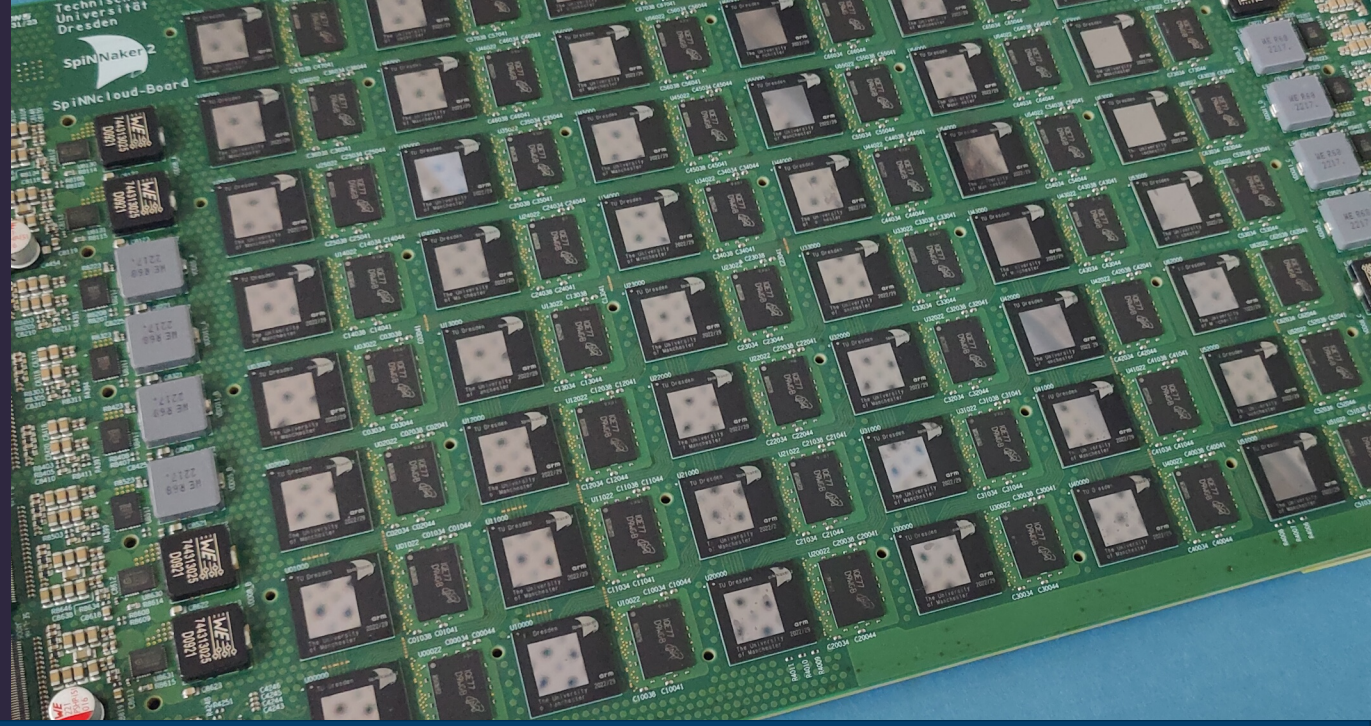
```
equations:
    kernel I_kernel = (e / tau_syn) * t * exp(-t / tau_syn)
    inline I_syn pA = convolve(I_kernel, spikes) * pA
    V_m' = -(V_m - E_L) / tau_m + (I_syn + I_e + I_stim) / C_m

parameters:
    C_m      pF = 250 pF      # Capacitance of the membrane
    tau_m    ms = 10 ms       # Membrane time constant
    tau_syn  ms = 2 ms        # Time constant of synaptic current
    refr_T   ms = 2 ms        # Duration of refractory period
    E_L      mV = -70 mV      # Resting potential
    V_reset  mV = -70 mV      # Reset potential of the membrane
    V_th     mV = -55 mV      # Spike threshold potential
    I_e      pA = 0 pA        # Constant external input current

input:
    spikes <- spike
    I_stim pA <- continuous

output:
    spike

update:
```



MODELING AND SIMULATING NEURON MODELS FROM NESTML ON NEUROMORPHIC HARDWARE

Charl Linssen <c.linssen@fz-juelich.de> | NEST Conference 2026 | June 16th, 2026

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Human Brain Project

JÜLICH Forschungszentrum
Member of the Helmholtz Association

Welcome!

T+00:00	Tutorial overview and accounts setup
T+00:10	Introduction to NESTML
T+00:20	Hands-on part 1: Generating code
T+00:30	Hands-on part 2: Single-neuron characterisation
T+00:40	Hands-on part 3: Balanced network

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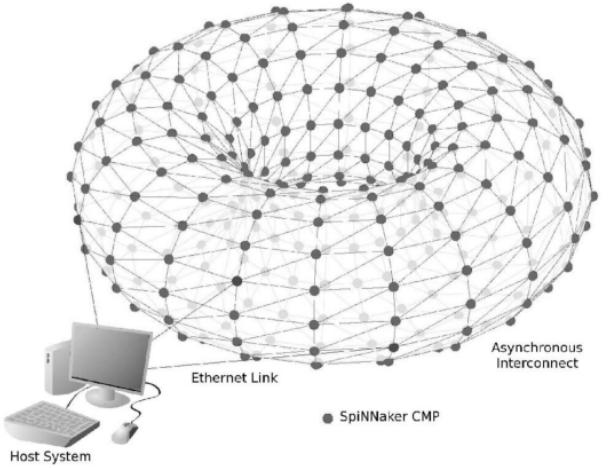
Introduction to SpiNNaker

SpiNNaker is a neuromorphic computing system designed to simulate large-scale spiking neural networks in real-time.

It features a massively parallel architecture comprising up to a million 32-bit ARM9 cores, interconnected through a toroidally wired, packet-based communication network, achieving high-speed, low-latency interprocessor communication.



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Running on EBRAINS

EBRAINS is an integrated digital brain simulation platform and provides cloud computing services for free.

To run the software used in this tutorial on EBRAINS, please create an account at:

https://ebrains.eu/sign-up

Running on the cloud

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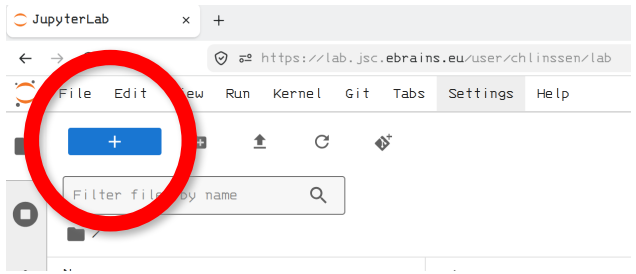
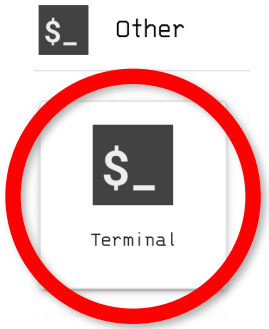
Where to find materials?

All contents of the tutorial (Jupyter notebooks) can be found on:

https://github.com/clinssen/ebrains-nestml-spinnaker

You can load these files onto your Jupyter instance by launching a new terminal:





And then typing the command:

```
git clone https://github.com/clinssen/ebrains-nestml-spinnaker
```

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Further reading

NESTML:

https://nestml.readthedocs.io/

SpiNNaker:

https://spinnakermanchester.github.io/